

Structural Vibration Analyzer

Current Hardware Setup Report

1 Introduction

The purpose of the current prototype is to create a low-cost structural vibration analyzer capable of collecting acceleration measurements from multiple locations on a structure.

The system uses **MPU-6050 accelerometers** to measure vibration, an **ESP32 DevKit** to control the sensors and collect the measurements, and an **I²C multiplexer** to allow several sensors to operate from the same ESP32.

The ESP32 does not perform the complete vibration analysis by itself. Its main purpose is to communicate with the sensors, collect the measurements at controlled time intervals, and transmit the data to a computer. The collected acceleration data can then be processed using Python or MATLAB to obtain acceleration-time plots and frequency spectra using the Fast Fourier Transform (FFT).

The basic hardware arrangement is:

Structure → **MPU-6050 Sensors** → **I²C Multiplexer** → **ESP32 DevKit** →
Computer → **FFT / Data Analysis**

2 Main Components

2.1 ESP32 DevKit

The ESP32 DevKit is used as the main controller of the vibration analyzer.

The ESP32 contains a 32-bit Xtensa LX6 processor and can operate at up to **240 MHz**. It also contains approximately **520 KB of SRAM**, which can be used to temporarily store sensor measurements before transmitting them to the computer.

The ESP32 Series also provides two I²C interfaces in addition to several other communication interfaces such as SPI and UART.

For this project, the important ESP32 functions are:

- Communicating with the I²C multiplexer.

- Selecting which MPU-6050 sensor is being read.
- Reading acceleration measurements.
- Maintaining approximately uniform sampling intervals.
- Temporarily storing sensor measurements.
- Sending measurements to the computer for analysis.

The ESP32 operates primarily with **3.3 V logic**, making it compatible with the sensor and multiplexer arrangement used in the analyzer.

The 240 MHz processor frequency should not be confused with the vibration measurement frequency. The processor is considerably faster than the sensors. Therefore, the main frequency limitation of the analyzer comes from the **MPU-6050 sensor and I²C communication**, rather than the ESP32 CPU itself.

3 MPU-6050 Motion Sensor

The MPU-6050 is a six-axis motion sensor containing:

- 3-axis accelerometer
- 3-axis gyroscope

For structural vibration measurements, the **accelerometer** is the main sensing element used.

The accelerometer measures acceleration independently in the:

- X direction
- Y direction
- Z direction

The MPU-6050 allows four selectable accelerometer ranges:

Accelerometer Range	Approximate Sensitivity
± 2 g	16,384 LSB/g
± 4 g	8,192 LSB/g
± 8 g	4,096 LSB/g
± 16 g	2,048 LSB/g

A smaller measurement range provides greater sensitivity, while a larger range allows stronger accelerations to be measured without saturating the sensor.

For relatively low-amplitude structural vibration, the **± 2 g or ± 4 g ranges** are generally more useful because they provide greater measurement sensitivity.

The MPU-6050 also communicates through I²C at speeds of up to **400 kHz**.

4 MPU-6050 Sampling Frequency

One of the most important limitations of the vibration analyzer is the sampling rate of the MPU-6050 accelerometer.

According to the MPU-6050 documentation, the accelerometer output rate is **1 kHz**.

This means that the accelerometer can produce up to approximately:

$$f_s = 1000 \text{ samples/second}$$

According to the Nyquist sampling principle, the theoretical highest frequency that could be represented from a 1 kHz sample rate is:

$$f_{Nyquist} = \frac{f_s}{2}$$

Therefore:

$$f_{Nyquist} = \frac{1000}{2} = 500 \text{ Hz}$$

The value of **500 Hz is only the theoretical Nyquist limit**. It does not mean that the MPU-6050 accurately measures vibration all the way to 500 Hz.

The sensor's internal bandwidth provides a more important practical limitation.

5 MPU-6050 Bandwidth

The MPU-6050 contains an internal **Digital Low-Pass Filter (DLPF)**. This filter can be configured to reduce high-frequency signals and noise before the data is used for analysis.

The accelerometer bandwidth settings specified by the manufacturer are:

DLPF Setting	Accelerometer Bandwidth	Approximate Delay
0	260 Hz	0 ms
1	184 Hz	2.0 ms
2	94 Hz	3.0 ms
3	44 Hz	4.9 ms
4	21 Hz	8.5 ms
5	10 Hz	13.8 ms
6	5 Hz	19.0 ms

The maximum listed accelerometer bandwidth is approximately:

$$\boxed{260 \text{ Hz}}$$

Therefore, although the 1 kHz sampling rate provides a theoretical Nyquist frequency of 500 Hz, the accelerometer’s own bandwidth limits the practically measurable vibration range to approximately **260 Hz or lower**.

This is important when describing the analyzer:

Sample Rate $\approx$ 1000 Hz

Nyquist Limit $\approx$ 500 Hz

Maximum MPU-6050 Accelerometer Bandwidth $\approx$ 260 Hz

The actual frequency range used during an experiment can be reduced further by selecting one of the lower DLPF bandwidths.

For example, if the expected structural vibration is below approximately 40 Hz, the **44 Hz DLPF setting** may be more appropriate than the 260 Hz setting because unwanted higher-frequency vibration and electronic noise can be reduced.

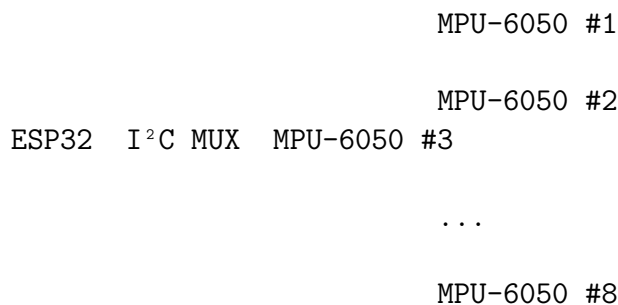
6 I²C Multiplexer

Multiple MPU-6050 sensors are connected to the ESP32 through an **8-channel I²C multiplexer**, such as the TCA9548A/PCA9548A.

The multiplexer provides one I²C connection to the ESP32 and divides it into **eight independent downstream I²C channels**.

This is necessary because several MPU-6050 sensors may use the same I²C address. Without the multiplexer, the ESP32 would not easily be able to distinguish between several sensors using identical addresses.

The arrangement is approximately:



The TCA9548A/PCA9548A can provide up to **eight separate channels**, allowing as many as eight same-address sensor devices to be connected through a single multiplexer.

The typical default I²C address of the multiplexer breakout is:

0x70

and it can be changed between **0x70** and **0x77** using its address selection pins.

7 I²C Communication Bandwidth

The PCA9548A multiplexer supports I²C clock frequencies from 0 Hz to:

400 kHz

The MPU-6050 also supports I²C communication at up to **400 kHz**.

Therefore, when these components are combined, the nominal maximum I²C bus clock for the sensor network is:

400 kHz

This 400 kHz value is the **communication clock**, not the vibration measurement frequency.

For example:

- **400 kHz I²C** = speed at which digital information travels between devices.
- **1 kHz sampling** = rate at which accelerometer measurements are generated.
- **260 Hz bandwidth** = approximate maximum accelerometer signal bandwidth.

These three values describe different parts of the system and should not be confused.

8 Effect of Using Multiple Sensors

The multiplexer does not read every sensor simultaneously. Instead, the ESP32 selects a multiplexer channel, reads that sensor, then switches to another channel.

For example, with three sensors:

```
Select Sensor 1
    ↓
Read X, Y, Z acceleration
    ↓
Select Sensor 2
    ↓
Read X, Y, Z acceleration
    ↓
Select Sensor 3
```

↓
 Read X, Y, Z acceleration
 ↓
 Repeat

Because all sensors share the same I²C communication path, increasing the number of sensors increases the amount of communication that must occur during each measurement cycle.

For this reason, although the MPU-6050 itself can provide a 1 kHz accelerometer output rate, the **actual achievable sampling rate per sensor in the complete multi-sensor analyzer must be experimentally verified.**

The limit will depend on factors including:

- Number of sensors connected.
- I²C clock frequency.
- Number of bytes read from each sensor.
- Multiplexer channel switching.
- ESP32 firmware execution time.
- Data transmission from the ESP32 to the computer.

This is one of the important limitations that can be investigated experimentally as part of the project.

9 Current Frequency Limits of the Analyzer

Based on the component specifications, the current system can be summarized as follows:

Parameter	Approximate Limit
ESP32 CPU Clock	240 MHz
I ² C Bus Clock	400 kHz
MPU-6050 Accelerometer Output Rate	1 kHz
Theoretical Nyquist Frequency	500 Hz
Maximum Accelerometer Bandwidth	260 Hz
Selectable DLPF Bandwidths	5–260 Hz
Accelerometer Measurement Range	±2, ±4, ±8, ±16 g
Multiplexer Channels	8

The most important distinction is that the analyzer should **not be described simply as a 500 Hz vibration analyzer.**

The theoretical 500 Hz Nyquist frequency results from the 1 kHz accelerometer sampling rate. However, the MPU-6050 accelerometer has a maximum specified bandwidth of approximately **260 Hz.**

A more appropriate preliminary description is therefore:

The present MPU-6050 based analyzer is intended primarily for low-frequency vibration measurements, with a sensor bandwidth configurable between approximately 5 Hz and 260 Hz. The accelerometer has a nominal output rate of 1 kHz, while the practical multi-sensor sampling rate remains to be experimentally determined.

10 Data Acquisition and Analysis

The complete measurement process is:

1. Mechanical vibration occurs in the structure.
2. Each MPU-6050 measures acceleration in the X, Y, and Z directions.
3. The ESP32 commands the multiplexer to select a sensor.
4. Acceleration measurements are read through the I²C bus.
5. The ESP32 repeats the process for the remaining connected sensors.
6. Measurements are transmitted to the computer.
7. The measurements are stored as acceleration versus time.
8. FFT analysis is performed to transform the acceleration measurements from the time domain into the frequency domain.
9. Peaks in the frequency spectrum can then be investigated as possible vibration or structural frequencies.

The current design therefore separates the system into two main sections:

Data Acquisition

MPU-6050 + Multiplexer + ESP32

Data Processing

Computer + Python/MATLAB + FFT

This approach allows the ESP32 to concentrate primarily on reliably collecting sensor measurements rather than performing all of the vibration calculations internally.

11 Current Limitations

At the present stage of development, several important limitations still need to be experimentally characterized.

Sensor Bandwidth

The MPU-6050 accelerometer has a maximum listed bandwidth of approximately **260 Hz**.

Sensor Sampling

The accelerometer operates with a 1 kHz output rate, but the usable sample rate when several sensors are being read sequentially must still be measured experimentally.

I²C Communication

Both the MPU-6050 and PCA9548A multiplexer are limited to approximately **400 kHz I²C operation**.

Multiple Sensors

Adding sensors increases the amount of data that must pass through the same I²C bus. Therefore, the maximum reliable sampling frequency is expected to decrease as more sensors are added.

Sensor Accuracy

The MPU-6050 is a low-cost MEMS sensor. Its noise, sensitivity, frequency response, mounting method, calibration, and repeatability must be experimentally investigated before determining what level of vibration measurement accuracy can be expected from the complete analyzer.

12 Conclusion

The current Structural Vibration Analyzer consists primarily of an **ESP32 DevKit, multiple MPU-6050 accelerometers, and an 8-channel I²C multiplexer**.

The ESP32 provides sufficient processing speed and memory for controlling the acquisition process, while the multiplexer makes it possible to connect multiple identical MPU-6050 sensors to the same controller.

The main frequency limitations currently come from the MPU-6050. The accelerometer provides a nominal output rate of **1 kHz** and configurable digital low-pass filter bandwidths ranging from approximately **5 Hz to 260 Hz**. The resulting theoretical Nyquist frequency is 500 Hz, although the sensor bandwidth limits the practical measurement range to approximately 260 Hz or below.

The next stage of the project is therefore not simply to rely on the manufacturer specifications, but to experimentally determine the **actual performance of the complete analyzer**, including maximum reliable sampling rate, usable frequency range, noise level, repeatability, and the effects of operating multiple sensors simultaneously.

These tests will establish the realistic operating limits of the prototype and determine its suitability for structural vibration measurements.